

# OpenTURNS: A Python package for uncertainty quantification

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## Software

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## Summary

OpenTURNS ([www.openturns.org](http://www.openturns.org)) is an open-source Python package for probabilistic modeling, uncertainty quantification (UQ), and machine learning. It provides a unified interface, granting users access to a comprehensive set of state-of-the-art algorithms and modeling tools. The library is supported by extensive, regularly updated easy to read documentation, including introductory examples, various use cases, tutorials, and user guides, enhancing its accessibility. OpenTURNS integrates seamlessly with NumPy, Pandas, and other scientific Python packages, allowing users to develop UQ workflows and generate reproducible results. Its modular architecture enables both newcomers and advanced users to access reliable, well-tested methods for uncertainty propagation, sensitivity analysis, and reliability assessment. Its active community provides additional support and resources through a Discourse forum ([OpenTURNS's Forum, 2020](#)) and a website. Stable releases are published on a semiannual basis, and the library is distributed through Conda, pip, and Debian repositories and available on Linux, Windows, and macOS.

## Statement of need

UQ plays a critical role in robust decision-making during the design and operation of complex systems. It encompasses probabilistic modeling (multivariate distributions, stochastic processes), sampling, functional algebra, uncertainty propagation, sensitivity analysis, reliability, optimization, inverse problems, and surrogate modeling. Validated and well-documented methods are essential for teaching, academic research, and industrial applications. However, many probabilistic modeling libraries are domain-specific—focusing on fields such as mechanical engineering, computational fluid dynamics, or hydrology—which limits the sharing of best practices. This underscores the need for a general-purpose, cross-disciplinary framework for probabilistic modeling.

## State of the field

There are many software packages and libraries for Uncertainty Quantification. Although the landscape is highly fragmented with significant specialization, these tools fall into two broad categories. First, there are general-purpose libraries, which aim to provide most of the features used in research and engineering to perform UQ studies. In this class of tools, we find Dakota ([Adams et al., 2025](#)), Raven ([Rabiti et al., 2017](#)), Sparse Grid++ ([Pflüger, 2010](#)), PSUADE ([Tong, 2024](#)), URANIE ([Blanchard et al., 2019](#)), UQLab ([Marelli & Sudret, 2014](#)), UQTK ([Debusschere et al., 2017](#)), OpenCOSSAN ([Patelli, 2016](#)), UQPy ([Olivier et al., 2020](#)), EasyVVUQ ([Suleimenova et al., 2021](#)), SmartUQ ([Absil et al., 2016](#)), and NESSUS

(McFarland & Riha, 2017), among others. These tools and libraries are provided in a variety of languages including Matlab (e.g., UQLab), Python (e.g., UQPy), and ROOT/C++ (e.g., URANIE). Some of these tools are open-source (e.g., EasyVVUQ), while others are commercial products (e.g., NESSUS). Other libraries have become open-source over time (e.g., UQLab became open-source in 2022). Many of these libraries are massive software projects (e.g., Dakota has more than 300,000 lines of code). Some of these tools have a graphical user interface (e.g., SmartUQ), but not all. The second category comprises a large number of specialized libraries. In this class of tools, we find, for example, GPy (GPy, 2012) for Gaussian Processes, QUESO (McDougall et al., 2017) for Bayesian inference, SALib (Herman & Usher, 2017) and sensitivity (Iooss et al., 2026) for sensitivity analysis and ChaosPy (Feinberg & Langtangen, 2015) for polynomial chaos expansion, among many others. Most of these tools are relatively small software projects (e.g., ChaosPy has approximately 30,000 lines of code). These specialized libraries are sometimes maintained by a small number of core developers, although some are based on many contributions (e.g., SALib has more than 40 contributors on GitHub). Furthermore, they are often easier to adapt for research purposes. This is the case, for example, for Python projects, which are easier for many users to extend compared to a C++ library. On the other hand, maintaining these libraries over time can be challenging.

Other Python packages exist for Monte Carlo sampling and other numerical methods (Virtanen et al., 2020), statistical modeling (Seabold & Perktold, 2010), machine learning (Pedregosa et al., 2011), UQ test functions (Wicaksono & Hecht, 2023), and quasi-Monte Carlo methods (P. Roy et al., 2023), among others. One of the goals of OpenTURNS is to provide a consistent interface for all these tasks, serving as a general-purpose tool for UQ.

Developing OpenTURNS as a standalone framework was necessary because integrating its specialized, tightly coupled C++ architecture for advanced reliability analysis and stochastic modeling into general-purpose libraries like SciPy or scikit-learn would have been architecturally incompatible and misaligned with their primary scopes. OpenTURNS was created in 2005 at a time when SciPy was a relatively new library (the development of SciPy was started in 2001) and scikit-learn did not exist (the first public release was in 2010). Today, this choice remains relevant as OpenTURNS is integrated into other C++ libraries such as Persalys, SALOME (see (SALOME Consortium, 2026)) and pSeven (see (pSeven SAS, 2026)), where the C++ implementation of OpenTURNS is mandatory.

## Software design

OpenTURNS relies on a dual-language architecture to balance computational efficiency with user accessibility. The core library is implemented in C++, which handles the memory management and heavy computational workloads required by Monte Carlo simulations, high-dimensional integration, and optimization algorithms. This C++ core is then exposed through a Python interface using SWIG.

The primary trade-off in this design is the increased complexity of maintaining a mixed-language codebase and managing bindings. However, this cost is outweighed by the performance gains in intensive numerical tasks.

From an architectural perspective, OpenTURNS is strictly object-oriented, mirroring mathematical concepts directly in the code. Users manipulate objects such as `Distribution`, `Process`, and `Function`. The `Function` class, for instance, provides an abstract representation of mathematical models, allowing for caching, composition, and automatic or finite-difference differentiation. This abstraction ensures that reliability and sensitivity algorithms can operate on complex industrial black-box models exactly as they would on simple analytical equations.

## 86 Features

87 OpenTURNS provides a wide range of easy to build multivariate probabilistic models: more  
88 than 60 continuous and discrete distributions, including extreme-value distributions, copula-  
89 based models (more than 15 copulas), and distribution transformations (such as truncation,  
90 conditioning, mixture, push-forward, and distribution algebra).

91 Multivariate distributions can be sampled using general sampling algorithms (Monte Carlo,  
92 Latin Hypercube Sampling (LHS), and optimized LHS) and quasi-Monte Carlo low-discrepancy  
93 sequences (such as Sobol' or Halton) for integration and model analysis.

94 OpenTURNS offers modeling features for stationary and non-stationary stochastic processes  
95 (e.g., ARMA processes and Gaussian processes in arbitrary dimensions) associated with several  
96 covariance kernels (e.g., exponential, Matérn, and squared exponential).

97 The library provides extensive data analysis capabilities. Dependency indicators can be computed  
98 (Pearson and Spearman correlations). Inference of parametric and non-parametric univariate or  
99 multivariate probabilistic models, including copula models, can be performed, along with several  
100 statistical tests to validate the estimation (quantitative tests such as the Kolmogorov-Smirnov  
101 or Lilliefors tests, and graphical tests such as the quantile-quantile plot).

102 The library provides an abstract representation of vector-valued or field-based mathematical  
103 functions that allows for composition, aggregation, symbolic or finite-difference derivatives,  
104 and caching. It also includes tools for wrapping black-box code that communicates through  
105 input/output file exchange.

106 Several sensitivity and reliability indices can be used to quantify the influence of one variable on  
107 another. Notable examples include Standardized Regression Coefficients (SRC), Sobol' indices,  
108 and Hilbert-Schmidt Independence Criterion (HSIC) indices. For the latter two, OpenTURNS  
109 provides the asymptotic distribution of the estimators, which allows the computation of  
110 confidence intervals.

111 OpenTURNS implements numerous reliability algorithms for rare-event probabilities. These  
112 include classic algorithms such as the First-Order Reliability Method, Second-Order Reliability  
113 Method, subset simulation, and other variance reduction methods. It also provides more  
114 recent methods such as Nonparametric Adaptive Importance Sampling (NAIS) and Importance  
115 Sampling by Cross-Entropy. An event can be a single occurrence (e.g., a threshold exceedance),  
116 a system event, or a condition based on a stochastic process.

117 For supervised learning, OpenTURNS enables the construction of surrogate models from data,  
118 such as linear regression models, polynomial chaos expansions, and Gaussian process regression.  
119 These surrogate models can also be built for models with functional inputs. The optimization  
120 can be performed on an arbitrary subset of the Gaussian process regression hyperparameters.

121 In many cases, probabilistic analyses use optimization algorithms (e.g., for supervised learning  
122 or calibration). OpenTURNS can be used to define a wide variety of optimization problems in  
123 a unified manner and provides users with solution algorithms from numerous existing libraries:  
124 NLOpt, IPOPT, pagmo, BONMIN, dlib, Ceres, cminpack, and COBYLA, among others. It  
125 can solve optimization problems with single or multiple objectives, with equality or inequality  
126 constraints, and with continuous or discrete variables.

127 Since many analyses require the computation of integrals, OpenTURNS provides integration  
128 methods based on sampling or quadrature: multivariate Gaussian quadrature (e.g.,  
129 Gauss-Legendre quadrature), Gauss-Kronrod, Smolyak, simplicial cubature, and other  
130 algorithms, including those from the Cuba library.

## Research impact statement

OpenTURNS is widely used in research, industrial engineering, and academic education. It has been integrated into university curricula (e.g., DTU Denmark) and engineering schools (e.g., ENSTA, ENPC, Centrale Nantes, among others) for courses on reliability analysis, risk assessment, and UQ.

The library has been introduced in several publications (e.g., Baudin et al., 2016) and used in a broad range of UQ studies across multiple science and engineering domains (see Dutfoy, 2023; Espoeys et al., 2025; Faure et al., 2024).

The Persalys graphical user interface (EDF & Phimeca Engineering, 2016), built on top of OpenTURNS, enables users to perform UQ analyses with no (or minimal) Python programming. OpenTURNS is a requirement for several other tools including (IRT Saint Exupéry, 2026) and (P. T. Roy et al., 2018) for example.

An annual Users' Day has been held since 2008, fostering collaboration and knowledge exchange among researchers, engineers, and educators.

## AI usage disclosure

Generative AI (Gemini 3.1 Pro) was used to verify the grammar and spelling of this paper, check the choices of the words and check that the paper comply with the requirements. No generative AI was used in the creation of the software, the technical documentation, or the algorithm design.

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## References

- Absil, C. O., Riccardi, A., Vasile, M., & Tardioli, C. (2016, March). SMART-UQ: uncertainty quantification toolbox for generalised intrusive and non intrusive polynomial algebra. *Proceedings of the 6th International Conference on Astrodynamics Tools and Techniques*. <https://indico.esa.int/event/111/contributions/238/>
- Adams, B. M., Bohnhoff, W. J., Bolstad, M. B., Dalbey, K. R., Ebeida, M. S., Eddy, J. P., Eldred, M. S., Hooper, R. W., Hough, P. D., Hu, K. T., Jakeman, J. D., Khalil, M., Maupin, K. A., Monschke, J. A., Prudencio, E. E., Ridgway, E. M., Rizzi, F., Robbe, P., Rushdi, A. A., ... Winokur, J. G. (2025). *Dakota 6.23.0 Documentation* (No. SAND2025-141790). Sandia National Laboratories. <http://snl-dakota.github.io>
- Baudin, M., Dutfoy, A., Iooss, B., & Popelin, A.-L. (2016). OpenTURNS: An Industrial Software for Uncertainty Quantification in Simulation. In R. Ghanem, D. Higdon, & H. Owhadi (Eds.), *Handbook of uncertainty quantification* (pp. 1–38). Springer International Publishing. [https://doi.org/10.1007/978-3-319-11259-6\\_64-1](https://doi.org/10.1007/978-3-319-11259-6_64-1)
- Blanchard, J.-B., Damblin, G., Martinez, J.-M., Arnaud, G., & Gaudier, F. (2019). The Uranie platform: an open-source software for optimisation, meta-modelling and uncertainty analysis. *EPJ Nuclear Sciences & Technologies*, 5, 4. <https://doi.org/10.1051/epjn/2018050>

- 174 Debusschere, B., Sargsyan, K., Safta, C., & Chowdhary, K. (2017). Uncertainty Quantification  
175 Toolkit (UQTK). In R. Ghanem, D. Higdon, & H. Owhadi (Eds.), *Handbook of uncertainty*  
176 *quantification* (pp. 1807–1827). Springer International Publishing. [https://doi.org/10.](https://doi.org/10.1007/978-3-319-12385-1_56)  
177 [1007/978-3-319-12385-1\\_56](https://doi.org/10.1007/978-3-319-12385-1_56)
- 178 Dutfoy, A. (2023). Uncertainty on estimated magnitudes: A new approach based on a  
179 Poisson point process of dimension 2. *Pure and Applied Geophysics*, 180(10), 919–933.  
180 <https://doi.org/10.1007/s00024-022-03222-6>
- 181 EDF, & Phimeca Engineering. (2016). *Persalys: Graphical User Interface for Uncertainty*  
182 *Quantification*. <https://www.persalys.fr/>.
- 183 Espoey, R., Brevault, L., Balesdent, M., Ricci, S., & Mycek, P. (2025). Multifidelity Bayesian  
184 Sequential Optimization and Reliability Assessment for Aerospace Systems Design. *Journal*  
185 *of Aerospace Information Systems*, 1–24. <https://doi.org/10.2514/1.1011614>
- 186 Faure, X., Lebrun, R., & Pasichnyi, O. (2024). Impact of time resolution on estimation of  
187 energy savings using a copula-based calibration in UBEM. *Energy and Buildings*, 311,  
188 114134. <https://doi.org/10.1016/j.enbuild.2024.114134>
- 189 Feinberg, J., & Langtangen, H. P. (2015). Chaospy: An open source tool for designing  
190 methods of uncertainty quantification. *Journal of Computational Science*, 11, 46–57.  
191 <https://doi.org/10.1016/j.jocs.2015.08.008>
- 192 GPy. (2012). *GPy: A Gaussian process framework in Python*. [http://github.com/SheffieldML/](http://github.com/SheffieldML/GPy)  
193 [GPy](http://github.com/SheffieldML/GPy).
- 194 Herman, J., & Usher, W. (2017). SALib: An open-source Python library for Sensitivity Analysis.  
195 *Journal of Open Source Software*, 2(9). <https://doi.org/10.21105/joss.00097>
- 196 Iooss, B., Da Veiga, S., Janon, A., & Pujol, G. (2026). *sensitivity: Global Sensitivity Analysis*  
197 *of Model Outputs*. <https://CRAN.R-project.org/package=sensitivity>
- 198 IRT Saint Exupéry. (2026). *GEMSEO: Generic Engine for Multidisciplinary Scenarios,*  
199 *Exploration and Optimization*. <https://gemseo.readthedocs.io/>
- 200 Marelli, S., & Sudret, B. (2014). UQLab: A Framework for Uncertainty Quantification in  
201 Matlab. *Proceedings of Vulnerability, Uncertainty, and Risk*, 2554–2563. [https://doi.org/](https://doi.org/10.1061/9780784413609.257)  
202 [10.1061/9780784413609.257](https://doi.org/10.1061/9780784413609.257)
- 203 McDougall, D., Malaya, N., & Moser, R. D. (2017). The Parallel C++ Statistical Library for  
204 Bayesian Inference: QUESO. In R. Ghanem, D. Higdon, & H. Owhadi (Eds.), *Handbook*  
205 *of uncertainty quantification* (pp. 1829–1865). Springer International Publishing. [https://doi.org/10.1007/978-3-319-12385-1\\_57](https://doi.org/10.1007/978-3-319-12385-1_57)
- 206
- 207 McFarland, J. M., & Riha, D. S. (2017). Probabilistic Analysis Using NESSUS (Numerical  
208 Evaluation of Stochastic Structures Under Stress). In R. Ghanem, D. Higdon, & H. Owhadi  
209 (Eds.), *Handbook of uncertainty quantification* (pp. 1733–1764). Springer International  
210 Publishing. [https://doi.org/10.1007/978-3-319-12385-1\\_54](https://doi.org/10.1007/978-3-319-12385-1_54)
- 211 Olivier, A., Giovanis, D. G., Aakash, B. S., Chauhan, M., Vandanapu, L., & Shields, M.  
212 D. (2020). UQpy: A general purpose Python package and development environment  
213 for uncertainty quantification. *Journal of Computational Science*, 47, 101204. <https://doi.org/10.1016/j.jocs.2020.101204>
- 214
- 215 *OpenTURNS's forum*. (2020). <https://openturns.discourse.group/>.
- 216 Patelli, E. (2016). COSSAN: A Multidisciplinary Software Suite for Uncertainty Quantification  
217 and Risk Management. In R. Ghanem, D. Higdon, & H. Owhadi (Eds.), *Handbook of*  
218 *uncertainty quantification* (pp. 1–69). Springer International Publishing. [https://doi.org/](https://doi.org/10.1007/978-3-319-11259-6_59-1)  
219 [10.1007/978-3-319-11259-6\\_59-1](https://doi.org/10.1007/978-3-319-11259-6_59-1)
- 220 Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel,

- 221 M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau,  
222 D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine Learning  
223 in Python. *Journal of Machine Learning Research*, 12(85), 2825–2830. [http://jmlr.org/  
224 papers/v12/pedregosa11a.html](http://jmlr.org/papers/v12/pedregosa11a.html)
- 225 Pflüger, D. (2010). *Spatially Adaptive Sparse Grids for High-Dimensional Problems*. Institut  
226 für Informatik, Technische Universität München; Verlag Dr. Hut. ISBN: 9783868535556
- 227 pSeven SAS. (2026). *pSeven: Design Space Exploration and Predictive Modeling Framework*.  
228 <https://www.pseven.io/product/pseven-core/>
- 229 Rabiti, C., Alfonsi, A., Cogliati, J., Mandelli, D., Kinoshita, R., Sen, S., Wang, C., & Chen,  
230 J. (2017). *RAVEN User Manual* (INL/EXT-15-34123). Idaho National Laboratory (INL).  
231 <https://doi.org/10.2172/1466671>
- 232 Roy, P. T., Ricci, S., Dupuis, R., Campet, R., Jouhaud, J.-C., & Fournier, C. (2018). BATMAN:  
233 Statistical analysis for expensive computer codes made easy. *Journal of Open Source  
234 Software*, 3(21), 493. <https://doi.org/10.21105/joss.00493>
- 235 Roy, P., Owen, A. B., Balandat, M., & Haberland, M. (2023). Quasi-Monte Carlo Methods in  
236 Python. *Journal of Open Source Software*, 8(84), 5309. [https://doi.org/10.21105/joss.  
237 05309](https://doi.org/10.21105/joss.05309)
- 238 SALOME Consortium. (2026). *SALOME: The Open Source Platform for Pre- and Post-  
239 Processing for Numerical Simulation*. <https://www.salome-platform.org/>
- 240 Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with  
241 Python. *Proceedings of the 9th Python in Science Conference*. [https://doi.org/10.25080/  
242 majora-92bf1922-011](https://doi.org/10.25080/majora-92bf1922-011)
- 243 Suleimenova, D., Arabnejad, H., Edeling, W. N., Coster, D., Luk, O. O., Lakhili, J., Jancauskas,  
244 V., Kulczewski, M., Veen, L., Ye, D., Zun, P., Krzhizhanovskaya, V., Hoekstra, A.,  
245 Crommelin, D., Coveney, P. V., & Groen, D. (2021). Tutorial applications for Verification,  
246 Validation and Uncertainty Quantification using VECMA toolkit. *Journal of Computational  
247 Science*, 53, 101402. <https://doi.org/10.1016/j.jocs.2021.101402>
- 248 Tong, C. H. (2024). *PSUADE: Problem Solving environment for Uncertainty Analysis and  
249 Design Exploration* (Version 3.0). <https://github.com/llnl/psuade>; GitHub.
- 250 Virtanen, P., Gommers, R., Oliphant, T. E., & others. (2020). SciPy 1.0: Fundamental  
251 Algorithms for Scientific Computing in Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- 252
- 253 Wicaksono, D., & Hecht, M. (2023). UQTestFuns: A Python3 library of uncertainty  
254 quantification (UQ) test functions. *Journal of Open Source Software*, 8(90), 5671.  
255 <https://doi.org/10.21105/joss.05671>